



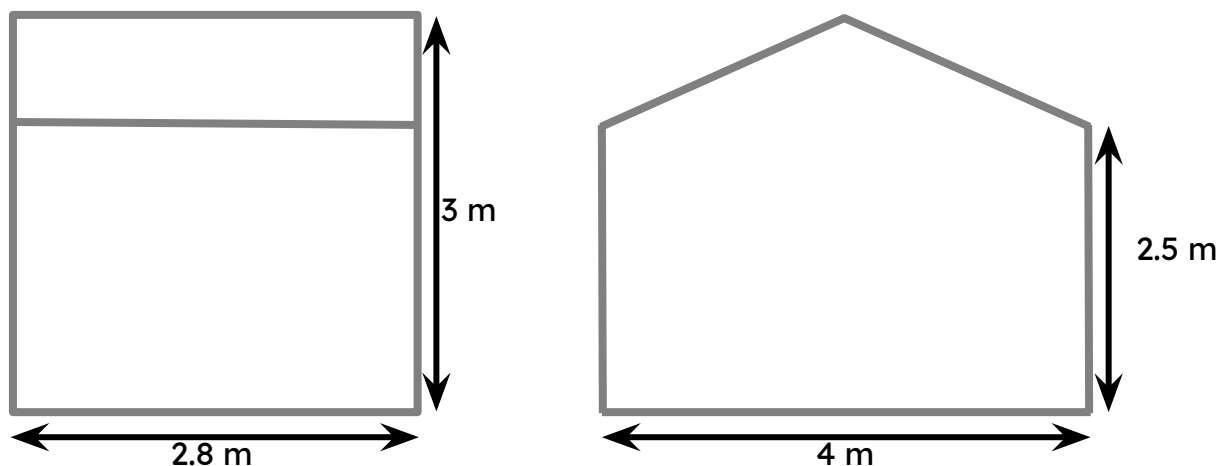
Construction

Task A: Using formulae in the construction industry

1) Here are the front and side elevations of a shed.

A builder needs to replace the rafters of the shed.

How long are the rafters?



2.062 m (3 d.p.)

2) $Q = A \times v$ where Q is the flow rate, A is the flow area and v is the velocity

Water runs through a cylindrical pipe at a velocity of 10 m/s.

The diameter of the pipe is 40 mm.

What is the flow rate in m^3 per hour?

$$Q = 0.02^2 \times \pi \times 10 \times 3600$$

$$Q = 45.24 \text{ m}^3 \text{ per hour (2 d.p.)}$$

3) $P = I \times V$ where P is the power (watts) , I is the current (amps) and V is the voltage (volts)

The voltage of electricity in UK households is 230 volts.

An electrician needs to install a 2.15 kW electric oven.

Circuits are available in 6, 10, 16, 20, 32, 40, 50 and 63 amps.

Which circuit should be installed?

$$I = 2150 \div 230$$

$$I = 9.35 \text{ amps (2 d.p.)}$$

A 10 amp circuit should be installed.

4) BTU = volume of the room $\times$ allowance for heat loss

Below are the dimensions of a bedroom.

Length = 4.6 m

Width = 2.6 m

Height = 2.4 m

a) Calculate the BTU.

$$4.6 \times 2.6 \times 2.4 \times 108 = 3100.032$$

b) The bedroom will have one radiator.

Which size should be installed?

$$600 \times 600$$

Room	Allowance
Bathroom	121.5
Bedroom	108
Dining room	135
Hallway	81
Kitchen	81
Living room	135

Height (mm)	Width (mm)	BTU
600	400	2184
600	600	3280
600	800	4372
600	1000	5464
600	1200	6556
600	1400	7649
600	1600	8744



Task B: Costing in the construction industry

1) This table shows the prices and lengths wood is sold in.

Length (mm)	Price
2400	£4.20
3000	£5.25
3600	£6.30
4200	£7.35
4800	£8.40

Calculate the cheapest price for purchasing the following:

a) Twenty 1 m lengths

The 3000 mm length produces three 1 m lengths.

6 × 3000 mm lengths + 1 × 2400 mm length will give twenty 1 m lengths at a cost of £35.70

b) Fifteen 1.8 m lengths

The 3600 mm length produces two 1.8 m lengths.

7 × 3600 mm lengths + 1 × 2400 mm length will give fifteen 1.8 m lengths at a cost of £48.30

c) 3 different ways to buy 600 1.2 m lengths

300 × 2400 mm length = £1260

200 × 3600 mm length = £1260

150 × 4800 mm length = £1260

d) Why are there 3 different ways for part (c)?

Because the 2400 mm, 3600 mm & 4800 mm lengths each produce no wastage and the cost per metre of wood is the same for all three.

